Name: ______________________ Problem Set 1 | Math 1180 | Cruz Godar

Due Thursday, September 10th at 11:59 PM

Complete the following problems and submit them as a PDF to Gradescope. You should show enough work that there is no question about the mathematical process used to obtain your answers, and so that your peers in the class could easily follow along. I encourage you to collaborate with your classmates, so long as you write up your solutions independently. Using AI tools to assist with the homework is not permitted.

Questions 1–5 concern the following graph of the function $f(x)$.

1. Find $f(-4)$, $f(-2)$, $f(0)$, $f(2)$, and $f(4)$.

2. Find $\lim_{x\to 0^+} f(x)$ and $\lim_{x\to 0^-} f(x)$.

3. Using interval notation, list all of the x-values where f is continuous and all of the x-values where f is discontinuous.

4. Find $\lim_{x\to \infty} f(x)$ and $\lim_{x\to -\infty} f(x)$.

5. Using interval notation, list all of the x-values where f is differentiable and all of the x-values where f is nondifferentiable.

In questions 6–8, find all of the critical points of the given function on the given interval, and classify them using the second derivative test. Then find the absolute maximum and minimum of the function.

6. $f(x) = 2x^3 - 3x^2 - 12x + 1$ on $[-2, 3]$.

7. $g(x) = x + \frac{1}{x}$ on $(0, 2]$.

8. $h(x) = \sin(x) + x$ on $[-2\pi, 2\pi]$.

In questions 9–11, find the derivative of the given function.

9. $p(t) = \dfrac{\tan(t)}{t^2 + 1}$.

10. $r(z) = \ln\left(z \cos(z)\right)$.

11. $l(k) = \sin(\cos(\tan(k)))$.